PE3101P: Decision Theory and Social Choice

Tutorial 5

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Throughout, an **option set** is a finite, non-empty set of available options, and a **choice function** C gives, for every option set S, a non-empty subset $C(S)$ of S: what may be chosen from S. A person's or voter's ranking of options is an ordering: complete and transitive.

Question 1

From any choice function C we can read off a ranking: $x \succsim y$ when $x \in C(\{x, y\})$. So $x \succ y$ when $C(\{x, y\}) = \{x\}$, and $x \sim y$ when $C(\{x, y\}) = \{x, y\}$. This ranking is complete, but it need not be transitive. An option is **maximal** in an option set, by a ranking, when nothing in the option set is strictly better than it. Sen's results relate consistency conditions on choice functions (Properties α, β and γ) to rankings. In this question we prove three facts about how they do so. Each part recalls the conditions it uses.

(a) In this part we will prove that if C satisfies expansion consistency (Property γ), then from every option set, C chooses every option that is maximal by the ranking read off from C. Recall:

> **Expansion consistency (Property γ)**: If x is chosen from an option set S_1, and x is chosen from an option set S_2, then x is chosen from the option set that combines them.
>
> $$[x \in C(S_1) \text{ and } x \in C(S_2)] \;\Rightarrow\; x \in C(S_1 \cup S_2)$$

Complete the proof. Fill each gap with one option set or the name of a condition.

Take any option set S, and any option x that is maximal in S. If $S = \{x\}$, then $C(S) = \{x\}$, since $C(S)$ is not empty. So suppose that S has other options too.

- Take any other option y in S. If x were not in $C(\{x, y\})$, then $C(\{x, y\})$ would be _____, since something is always chosen from $\{x, y\}$; and then $y \succ x$, which is false, since x is maximal in S. So $x \in C(\{x, y\})$.

- Combining the option sets $\{x, y\}$, for every other option y in S, gives the option set _____. So $x \in C(S)$, by _____ (applied again and again).

(b) In this part, $\succsim$ is a complete and transitive ranking, and C chooses, from every option set, the maximal options by $\succsim$. We will show that C satisfies Property β:

> **Property β**: If x and y are both chosen from T, and $T \subset S$, then y is chosen from S if and only if x is.
>
> $$[x, y \in C(T) \text{ and } T \subset S] \;\Rightarrow\; [y \in C(S) \Leftrightarrow x \in C(S)]$$

Take option sets T and S with $T \subset S$, and options x and y that are both in $C(T)$. Suppose that $y \in C(S)$.

(i) Using the completeness of $\succsim$, show that $x \succsim y$.

Recall:

> **Transitivity**: If $s \succsim t$ and $t \succsim u$, then $s \succsim u$. (So also: if $s \succ t$ and $t \succsim u$, then $s \succ u$.)

(ii) Show that nothing in S is strictly better than x.

So $x \in C(S)$. The same argument with x and y swapped shows that if $x \in C(S)$, then $y \in C(S)$. So C satisfies Property β.

(c) In this part, C is any choice function that satisfies Property β (as in (b)) and contraction consistency (Property α), and $\succsim$ is the ranking read off from C. We will prove that indifference ($\sim$) by this ranking is transitive. Recall:

> **Contraction consistency (Property α)**: If x is chosen from an option set S, and x is in a smaller option set T inside S, then x is chosen from T.
>
> $$[x \in C(S) \text{ and } x \in T \subset S] \;\Rightarrow\; x \in C(T)$$

Complete the proof. Fill each gap with one option set or the name of a condition.

Take any three different options x, y and z with $x \sim y$ and $y \sim z$. So $C(\{x, y\}) = \{x, y\}$ and $C(\{y, z\}) = \{y, z\}$. Let $S = \{x, y, z\}$.

- x and y are both chosen from $\{x, y\}$, which is inside S. So if either of them is in $C(S)$, both are, by _____. Likewise for y and z.
- So, since $C(S)$ is not empty, $C(S) =$ _____.
- So x and z are both chosen from $\{x, z\}$, by _____. So $x \sim z$.

Question 2

When two people are decisive over the same pair and disagree about it, each excludes one of its two states, so rights conflict even without Pareto. This question asks what else such a conflict needs. Recall:

> **Decisiveness**: A person is decisive over a pair of social states when, whichever of the two they strictly prefer, the other is not chosen from any option set containing both.

Say that a state in an option set is **excluded** when someone's decisiveness or right means that it is not chosen from that option set.

(a) Suppose that person 1 is decisive over one pair of social states, and person 2 over a different pair (the two pairs may share a state). Leave Pareto aside. Is there an option set, and a ranking for each person, under which every state in the option set is excluded by someone's decisiveness? If so, give one. If not, say why not.

Gibbard gives each person a right over their own features instead, and rights of this kind can conflict without Pareto. Two people, 1 and 2, each choose whether to keep their own hair long (L) or short (S). A state is written with person 1's hair first: for example, SL is the state in which person 1's hair is short and person 2's is long. Each person ranks the four states LL, LS, SL and SS from best to worst, with no ties, and has a right over their own hair:

Rights over own hair: If two states differ only in one person's hair, the one of the two that this person prefers excludes the other: the other is not chosen from any option set containing both.

So each person's right covers two pairs, not one: person 1's covers $\{\text{LL}, \text{SL}\}$ and $\{\text{LS}, \text{SS}\}$, and person 2's covers $\{\text{LL}, \text{LS}\}$ and $\{\text{SL}, \text{SS}\}$. Say that a person's preference about their own hair is **unconditional** when they prefer the same length for themselves whatever the other person's hair is. For example, person 1's preference is unconditional, with long hair preferred, when person 1 prefers LL to SL and also prefers LS to SS.

(b) In this part we show that one unconditional preference is enough to prevent a conflict. Suppose that person 1's preference about their own hair is unconditional; person 2's need not be. Show that from every option set whose states are among these four, at least one state is excluded by no right. (Start with the states of the option set in which person 1's hair is the length person 1 prefers.)

Question 3

A course (PE3101P) has one place left. Rosie and Jim may each apply for it. Rosie decides first; Jim sees what she decided, and then decides. The result is one of three social states:

Choices	State	What happens
Rosie applies (whatever Jim does)	y	Rosie takes the place
Rosie does not apply, and Jim applies	z	Jim takes the place
Neither applies	x	Nobody takes the place

Whether Rosie applies is a personal matter for her, and whether Jim applies is a personal matter for him. Say that a person has a **right** to a feature of the result when some choice of theirs guarantees that feature, whatever the other person chooses.

(a) Fill in the table. In each cell, write "no" if the person has no right to the feature; if they have one, write the choice that guarantees it.

	They take the place themselves	They do not take the place
Rosie		
Jim		

Rosie does not really want the course, but she would rather take it herself than see Jim take it, since she doesn't want him to find out about contraction consistency. Jim wants the place most; failing that, he would rather it stayed empty than went to Rosie. So their rankings are:

	Rosie	Jim
Best	Nobody (x)	Jim (z)
Middle	Rosie (y)	Nobody (x)
Worst	Jim (z)	Rosie (y)

(b) Each of them knows the other's ranking, and each chooses what is best for themselves. Which state results?

(c) Suppose instead that Jim cannot apply, so that only Rosie decides. Which state results?

Think of the state that results as what is chosen from the states that the procedure can lead to: from $\{x, y, z\}$ in (b), and from $\{x, y\}$ in (c). Recall:

> **Contraction consistency (Property α)**: If x is chosen from an option set S, and x is in a smaller option set T inside S, then x is chosen from T.

(d) Does this way of choosing satisfy contraction consistency (Property α)? If not, give the states and option sets that show it.

On Sen's way of stating rights, Rosie is decisive over $\{x, y\}$, since these two states differ only in whether she applies. Recall:

> **Decisiveness**: A person is decisive over a pair of social states when, whichever of the two they strictly prefer, the other is not chosen from any option set containing both.

(e) Does the result in (b) respect Rosie's decisiveness over $\{x, y\}$?

(f) On the same way of stating rights, is Jim decisive over any pair of social states? If so, which? If not, say why not.

(g) Now return to rights as choices. In (b), did either person's choice stop the other from getting a feature that, by your table in (a), the other had a right to?

Recall:

> **Pareto**: If everyone strictly prefers s to t, then t is not chosen from any option set containing s.

(h) Does the result in (b) satisfy Pareto?

Question 4

Recall that an ordering can be written as a list of **levels**, from top to bottom: options in the same level are tied, and an option in a higher level is strictly preferred to one in a lower level. A **ranking shift** takes one option out of its level and puts it back somewhere else in the list, as a new level of its own or into another level. Everything else stays as it was: every pair of options not involving the moved option is ranked as before.

(a) A voter's ordering is $a \succ b \sim c \succ d$, the list $\{a\}$, $\{b, c\}$, $\{d\}$. For each ordering below, can it be obtained from this one by a single ranking shift of b? Write "yes" or "no".

	Ordering	By a ranking shift of b?
(i)	$a \sim b \succ c \succ d$	
(ii)	$a \succ c \succ b \sim d$	
(iii)	$a \succ b \succ d \succ c$	
(iv)	$a \sim b \sim c \succ d$	

In the rest of this question, there are three voters and four options, a, b, c and d, and every voter's ordering is strict (no ties). A social ordering function gives a social ordering $\succsim$, complete

and transitive, for each profile of voters' orderings. The one here satisfies Unrestricted Domain and

Pareto: If every voter prefers x to y, then $x \succ y$.

IIA: Take any two profiles, and any two options x and y. If every voter ranks x against y the same way in both profiles, then society ranks x against y the same way in both.

Here are two profiles. P' is made from P by ranking shifts of b by voters 1 and 2; voter 3's ordering is the same in both.

P:

Voter	Ordering
1	$b \succ a \succ c \succ d$
2	$c \succ d \succ a \succ b$
3	$c \succ b \succ d \succ a$

P':

Voter	Ordering
1	$a \succ b \succ c \succ d$
2	$c \succ b \succ d \succ a$
3	$c \succ b \succ d \succ a$

At P, society's ordering is $c \succ d \succ a \succ b$. We work out society's ordering at P', one pair at a time.

(b) (i) For which pairs of options does IIA carry society's ranking over from P to P'?

(ii) For each of the other pairs, which condition, or which property of a social ordering, fixes society's ranking at P'?

(iii) So what is society's ordering at P'?

Question 5

This question walks through the last stage of the proof of Arrow's Theorem, with three voters, 1, 2 and 3, and three options, a, b and c. Part (a) finds the pivotal voter for b. Part (b) shows, for any social ordering function satisfying the conditions below, that the pivotal voters for the other two options must be the same voter as the pivotal voter for b. Part (c) returns to the function in (a) and shows that its pivotal voter for b is a dictator.

Throughout, a social ordering function gives a social ordering $\succsim$, complete and transitive, for each profile, and satisfies Unrestricted Domain and

Pareto: If every voter prefers x to y, then $x \succ y$.

IIA: Take any two profiles, and any two options x and y. If every voter ranks x against y the same way in both profiles, then society ranks x against y the same way in both.

To find the **pivotal voter** for an option, start from a profile in which every voter puts that option strictly at the bottom. Let the voters, one at a time and in some fixed order, each move that option to the top of their own ranking, changing nothing else. The pivotal voter is the voter whose move first makes society put the option strictly at the top.

(a) Here the voters move b to the top of their rankings in the order voter 3, then voter 1, then voter 2:

Profile	Voter 1	Voter 2	Voter 3	
P_0	$a \succ c \succ b$	$c \succ a \succ b$	$a \succ c \succ b$	
P_1	$a \succ c \succ b$	$c \succ a \succ b$	$b \succ a \succ c$	(voter 3 has moved b to the top)
P_2	$b \succ a \succ c$	$c \succ a \succ b$	$b \succ a \succ c$	(voter 1 has moved b to the top)
P_3	$b \succ a \succ c$	$b \succ c \succ a$	$b \succ a \succ c$	(voter 2 has moved b to the top)

At P_1, society's ordering is $a \succ c \succ b$, and at P_2 it is $b \succ a \succ c$. Which voter is the pivotal voter for b?

Recall:

Decisiveness: A voter i is decisive over a pair $\{x, y\}$ when, at every profile, whichever of the two they prefer, society prefers too (where $x \succ_i y$ means that voter i strictly prefers x to y): if $x \succ_i y$, then $x \succ y$; and if $y \succ_i x$, then $y \succ x$.

Pivotal voters are decisive: For any social ordering function satisfying Unrestricted Domain, Pareto and IIA, with finitely many voters and at least three options, the pivotal voter for an option, found in this way, is decisive over every pair of options not involving that option.

Write $V(a)$, $V(b)$ and $V(c)$ for the pivotal voters for a, b and c. So $V(a)$ is decisive over $\{b, c\}$, $V(b)$ over $\{a, c\}$, and $V(c)$ over $\{a, b\}$.

(b) In this part, take any social ordering function satisfying Unrestricted Domain, Pareto and IIA whose pivotal voter for b is voter 2. Nothing is known about who $V(a)$ and $V(c)$ are.

(i) Suppose, for contradiction, that $V(a)$ is not voter 2. Give a single profile that shows this is impossible, whichever of voters 1 and 3 is $V(a)$, and whoever $V(c)$ is: at your profile, the decisiveness of $V(a)$, $V(b)$ and $V(c)$ must force society's ranking not to be transitive. Show how society must rank each pair, and say whose decisiveness forces it.

Voter	Ordering
1	
2	
3	

Pair	Society	Forced by
$\{a, b\}$		
$\{a, c\}$		
$\{b, c\}$		

(ii) By (i), $V(a)$ is voter 2. Suppose, for contradiction, that $V(c)$ is not voter 2. Does your profile from (i) also show that this is impossible? If it does, show how society must rank each pair at it, and say whose decisiveness forces it. If not, say why not, and give a profile that does.

(c) Return to the social ordering function in (a).

(i) Which voter is its dictator (a voter who is decisive over every pair)? Give your reason, using (a) and the argument of (b).

(ii) At the profile R below, for which pairs of options does the dictator's decisiveness settle how society ranks them? For any pair it does not settle, say why not.

R:

Voter	Ordering
1	$a \sim b \succ c$
2	$b \succ c \succ a$
3	$c \succ a \succ b$

(iii) Suppose the voters had instead moved b to the top in the order voter 1, then voter 2, then voter 3, starting from P_0. Which voter would be the pivotal voter for b?